YUPU YAO

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EDUCATION

University of Electronic Science and Technology of China
B.Eng. in Communication Engineering *Sept 2022 - June 2026*

WORK EXPERIENCE

Pennsylvania State University *Jun 2026 – Present*
Research Assistant(Remote) | Advised by Dr.Lu Cheng *Pennsylvania, USA*

- Conducting research on representation learning and latent reasoning, investigating the relationship between learned representation structures and reasoning, planning, and generalization.
- Independently initiated two research directions focusing on the mathematical foundations and computational dynamics of learned representations.
- Contributing to collaborative research on latent structure alignment for world model planning.

Infinigence AI *Aug 2025 – Mar 2026*
Inference Framework Developer Intern *Shanghai, China*

- Led the development of an SGLang- and vLLM-based heterogeneous inference framework, designing a GPU-FPGA scheduler to address distributed attention and MoE bottlenecks through computation-communication overlap.
- Implemented Prefill/Decode disaggregation to reduce interference between inference stages and improve overall throughput.
- Deployed and optimized large-scale models (DeepSeek 671B, Qwen 235B) with FP8/INT4 support on distributed 8xH20 and 8xA100 GPU clusters.

RESEARCH EXPERIENCE

Relational Knowledge in Looped Transformers *Sept 2026 – Present*

- Investigating how relational knowledge is represented, transformed, and reused across the layers and iterations of looped Transformers.
- Exploring representation-level mechanisms underlying multi-step reasoning and the generalization of learned relational knowledge.
- Analyzing internal representations and computational dynamics to understand how learned knowledge can support more effective reasoning.

Beyond Finite-Dimensional Representations in Linear Space *Sept 2026 – Present*

- Investigating the role of representation spaces in planning, with an emphasis on the relationship between latent structure and computational capabilities.
- Exploring mathematical perspectives on representation learning beyond conventional finite-dimensional formulations.
- Studying how the structure of representation spaces may influence planning, generalization, and the reuse of learned knowledge.

Deconstructing Attention Mechanisms via Implicit Hilbert Ladder *Feb 2024 - June 2024*

- Proposed and implemented the Implicit Hilbert Ladder (IHL) framework, modeling Multi-Head Self-Attention (MSA) from a Reproducing Kernel Hilbert Space (RKHS) perspective to provide a new theoretical understanding of Transformers.
- Developed a novel mechanism, Hilbertian Attention Cascade (HAC), replacing self-attention with kernel mappings in RKHS. Achieved a >5% accuracy increase in early-stage training on CIFAR-10, with 16% fewer parameters and a 12.3% reduction in GFLOPs.
- Validated the theoretical advantages of IHL via Rademacher complexity generalization bounds, Neural Tangent Kernel (NTK) convergence proofs, and gradient flow analysis, establishing a theoretically-grounded attention alternative.

Resource Allocation for Semantic-Aware Communication Networks *Aug 2023 - Feb 2024*

- Deployed and trained a Deep Q-Network (DQN)-based intelligent resource allocation algorithm, improving average user Quality of Experience (QoE) by 23.5% and system convergence speed by 18% in simulated wireless networks.
- Proposed a distributed two-tier control strategy integrating local DQNs with a global coordination module, reducing training communication overhead by 28.6% while maintaining near-optimal global performance.
- Effectively addressed the joint task of beamforming, power control, and semantic resource allocation, demonstrating a 19% improvement in average policy robustness under multi-task, high-dimensional state spaces.

PROJECTS

Development of a Multifunctional Autonomous Navigation Drone *Mar 2025 - June 2025*

- Designed and deployed a fully autonomous UAV with real-time perception and planning capabilities. Integrated PX4 with Orange Pi, VINS-MONO for localization, Metric3D for depth estimation, YOLOv8n for detection, and EGO-Planner for trajectory planning. Achieved full ROS 2 system integration with low-level refactoring for performance optimization.

PUBLICATIONS AND MANUSCRIPTS

[1]Ke Fang*, **Yupu Yao***, Lu Cheng. ATLAS: Aligned Transport of Latent Structure for Reliable World Model Planning. *Preprint*

[2]**Yupu Yao** . Generalize neural network through smooth hypothesis function. *ICLR 2024(Tiny)*

[3]**Yupu Yao**, Bowen Yang. GeoPE: A Unified Geometric Positional Embedding for Structured Tensors. *Preprint*

[4]**Yupu Yao**, Shangqi Deng, Zihan Cao, Harry Zhang, Liang-jian Deng. APLA: Additional perturbation for latent noise with adversarial training enables consistency. *Preprint*

SERVICE

Conference Reviewer
ICLR 2027; WACV 2027 Workshop on WAM-AD; AAAI 2027 workshop on SAFE
Journal Reviewer
IEEE Internet of Things Journal (IoT)

TECHNICAL SKILLS

- **ML/DL Frameworks:** Pytorch, Tensorflow, JAX, Triton
- **Training Frameworks:** DeepSpeed
- **Inference Frameworks:** SGLang, vLLM